

G57 Twist Brackets: Adaptive All-g57 Bracket Insertion for `fmix`, and the Measurements That Validate It

local_mixing working notes

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Abstract

The `fmix` conjugation twist used to bracket its window with two 3-CNOT swap packets — six `comp=0` single-control gates per twist, a distinctive syntactic fingerprint that the frozen store cannot digest. This note describes the replacement (`-twist-g57`, commits `4e9dbf40` and `83ae0d11`): each bracket is spelled as a shortest *all-g57 word* solved online per seam, sited adaptively so it absorbs neighborhood gates — the hidden-SAMF mechanism of `ssg`, made XGate-native. We describe the word engine, the two-generation placer, and the measurement campaign: A/B/C-schedule mixing runs at two twist rates against no-twist and legacy-twist baselines, a placement factorial, an ancestry-instrument audit, and a functional-legibility audit (degree-1/degree-2 affine plates and the stress battery). Headline: at rate 0.002 the new brackets give $\sim 40\%$ smaller circuits and $5\text{--}7\times$ the ancestry transport of the legacy packets; and across *all* twist variants and rates, no functional instrument in the battery detects either harm or benefit from twisting at these doses — the one measurable twist effect is syntactic (write-census smearing).

1 The bracket problem

A twist rewrites $W \mapsto P(PWP)P \equiv W$: it conjugates a window’s interior in place and pays for two bracket packets P . With $P = \text{swap}(a, b)$ realized as 3 CNOTs per side, every twist deposits six `comp=0` single-control gates. Three problems: (i) the packets are a syntactic fingerprint (nothing else in the mixed bulk looks like them); (ii) the frozen store’s match rate collapses on CNOT-bearing windows (the $m=6$ cliff), so the packets are indigestible and accumulate; (iii) they are pure cost — placed uniformly at random, nothing ever cancels.

The fix follows the `ssg` hidden-SAMF idea: spell the bracket in the bulk material’s own alphabet (g57 gates, `comp=1` width-2), and *site* it where the neighborhood pays part of the bill.

2 The swap-word engine

The bracket seam only needs the shortest all-g57 word for a *permutation*: $R \equiv \text{ctx} \cdot S$ at the opening seam, $S \cdot \text{ctx}$ at the closing seam, where ctx is a few real neighborhood gates and $S = \text{swap}(a, b)$. This is answerable online, with no precomputed replacement database: on 4 abstract wires a 16-state permutation packs into one `u64`; the radius-4 BFS ball over the 24 g57 gates (165,443 permutations) builds once in ~ 10 ms, and a meet-in-the-middle scan answers any target up to word length 7 in $\sim 200\text{--}350\ \mu\text{s}$. The tables are target-independent: one engine serves every context shape, any consumption depth, 3-wire supports, non-g57 neighbors, and (later) the negation twist variants. An earlier design — a SQLite enumeration of 218k syntactic contexts requiring offline synthesis — was

retired in favor of this after its $\leq k+3$ replacement contract proved unachievable at $k=2$ in its own scope (every such context sits at distance 6).

Ground facts (exhaustive; Appendix A for the full tables): $\text{dist}(S) = 6$ over the g57 alphabet on 3 *and* 4 wires — the fourth wire buys nothing; there are exactly **64** minimal words (32 per choice of helper wire, none genuinely 4-wire). Consuming k neighborhood gates nets $6 - 2k$ when they cancel: $k=1$ always possible in the abstract (8 of 24 gates are attachable for fixed S), $k=2$ net +2, $k=3$ net 0 (48% of same-support triples), and with *non-g57* context (worth several g57s each) a seam can net *negative* — a twist that shrinks the circuit.

3 The placer

v1 (anchor-first). Candidate wire pairs (a, b) are seeded from the boundary gates’ own pins (a uniformly random b almost never lands inside the 4-wire seam support), plus one uniform pair to keep fresh-wire routing on the menu. Both seams are solved for every candidate; cheapest total net wins. Each seam may consume up to 3 context gates of any shape. Every splice is verified against the reference 3-CNOT packet under local verification.

v1’s failure mode. The per-seam net histogram showed almost exactly *one* bare (+6) seam per twist: the closing bracket inherits (a, b) from the opening seam, and its fixed neighbor almost never pins both wires.

v2 (slide + joint acceptance). Two composable fixes, each with an env kill-switch for A/Bs (TWIST_G57_NO_SLIDE, TWIST_G57_NO_RETRY; both default on):

- *Slide*: a seam that stays bare walks its bracket outward (up to 512 gates, extending the conjugated window over what it passes — window length was a random draw, and a relabel is far cheaper than the word the slide saves) to the next g57 pinning both twist wires.
- *Joint acceptance*: both ends are solved before committing; a window whose best plan nets worse than +8 in total is redrawn (up to 4 tries), so one side stays bare only when its partner’s match pays for it (net $\leq +2$) or every redraw failed.

Always-advance. Every inserted gate takes the ballistic birth-advance unconditionally (the **-db-advance** treatment, aimed outward per bracket) — bracket material rides away from the seam instead of sitting on the window edge. This is independent of the **-db-advance** flag itself, which still governs DB splice products only.

Ancestry. Consumed context unions its litters’ ancestor sets into the replacement litter, exactly as a DB splice would (v1 dropped them — an instrument bug worth +1.5–10% of **anc**, isolated by rerunning v1-placement arms with only the recording changed).

4 Mixing experiments

All runs: the same 20,000-gate, 64-wire sample (the first 3/7 of a Gray-fold gadgetized sliced sandwich, $n=16$), seed 101, 2M moves, ancestry recording on, C schedule (**-p-mix 0.1**) unless noted; “legacy” is the shipped swap-family 3-CNOT packet twist, rates are **-p-twist**.

At matched *effective work* (cumulative moves per gate), the ancestry cost of twisting is: legacy –89% vs. no-twist, g57 –45% — the new brackets keep 5–6 $\times$ more transport at equal dose, *and* end smaller, despite emitting more gates per twist (~ 8 –9 net vs. 6): the brackets are store-digestible, so COMP chews them back down instead of accumulating CNOT bulk.

C-schedule arm	size	anc	span	polf
no twist	48,493	7,941	8,626	0.000
legacy 0.002	133,665	425	1,111	0.422
legacy 0.01	262,115	62	369	0.487
g57 v1 0.002	86,696	2,604	3,374	0.000
g57 v1 0.01	216,281	637	1,341	0.000
g57 v2 0.002	78,233	3,038	3,895	0.000
g57 v2 0.01	168,192	921	1,625	0.000

Table 1: Final size and ancestry measures, C arm. The A and B schedules give the same ordering. **polf**= 0 because the g57 path is pure swap — the negation arms (the polarity-mixing capability) still use the legacy packet; their word tables are the open extension.

Placement factorial (C arm, rate 0.01): mean net gates per seam 4.51 (v1) $\rightarrow$ 4.38 (slide only) $\rightarrow$ 3.93 (retry only) $\rightarrow$ **3.72** (both); bare seams 50% $\rightarrow$ 35%; size 216k $\rightarrow$ 168k; anc 637 $\rightarrow$ 921. The joint-acceptance retry is the larger lever; the slide composes. Cost: $2.6\times$ the MITM solves and ~ 1.6 extra window draws per twist — microseconds per move at production rates.

5 Is the ancestry drop a mixing loss? The instrument audit

Twists still cost **anc**/**span** relative to no-twist even under v2. Two audits asked whether that reading is real.

(1) Recording audit. A conditional census over all run states (parsing litters and ancestor sets) shows the reported **anc** was *already* conditional on non-empty sets, and the empty-ancestry gate fraction is moderate (3–21% for g57, up to 46% for legacy at 0.01) — so the drop is not mean-deflation by ancestry-less brackets; conditioning changes nothing about the ordering. The union-recording fix (above) moves **anc** by only +1.5–10%.

(2) Functional audit. Affine-reconstruction plates (**hmap_affine**) of the *input circuit's* prefix progress against each mixed circuit, read by the ridge measure:

- Degree 1: identical for all seven arms ($\rho = 1.000$, perm $z \approx 7.1$, depth 0.22–0.26). *Caveat discovered in the process:* every current twist type is an affine conjugation and the degree-1 measure is affine-invariant — it is **blind to twist effects by construction**, in both directions.
- Degree 2 (2,081 regressors — not affine-blind): the diagonal is at ceiling in every arm ($\rho = 1.000$, $z \approx 4.5$); depth 0.305 (no twist) vs. 0.26–0.30 (twisted) — at most a marginal hint of benefit, within noise.
- Stress battery (bias-sensitive statistical probes; single-wire / XOR-pair predictors on state bits, parities, and trajectory bits): no benefit — prominence 0.26–0.31 above floor in *every* arm including no-twist, with progress monotonicity $\rho = 0.85$ –0.99, $z = 7$ –9 everywhere. The progress clock is readable by near-trivial predictors in all arms; no twist dose or variant dents it.
- The one measurable twist effect is *syntactic*: the per-wire write-count bimodality gap falls from 0.283 (no twist) to 0.09–0.22 under twists.

Verdict. At these doses, twisting costs real ancestry transport and size while buying nothing measurable on functional channels; its measurable value is syntactic-census smearing. The ancestry

fmix A/B/C $\times$ twist rate — legacy swap family (---) vs --twist-g57 v2: slide+joint-retry+anc-union (-.-.-)

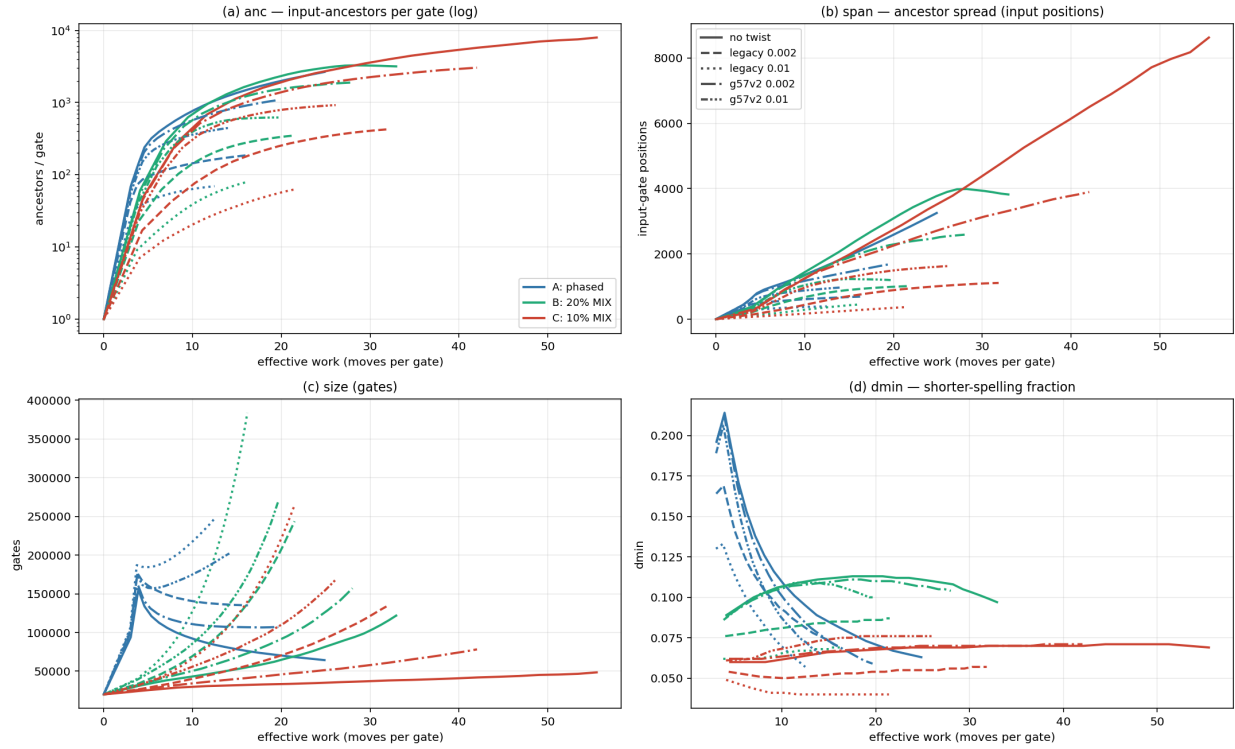


Figure 1: A/B/C $\times$ twist rate: legacy packets (dashed/dotted) vs. g57 v2 brackets (dash-dot styles), no-twist solid. Panels: anc, span, size, dmin vs. effective work.

gap *between twist variants* is therefore not a mixing-quality gap — but the case for twisting at all now rests on adversary classes not yet in the battery. Practical recommendation: keep `-p-twist` at 0.002 or below, use `-twist-g57` so the dose costs 5–6 $\times$ less, and treat any proposal to raise twist rates as requiring a demonstrated adversary it defeats.

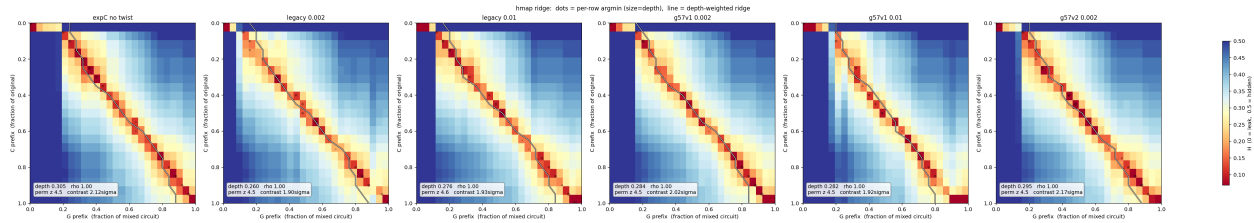


Figure 2: Degree-2 plates, six of the seven C-family arms: the input-progress diagonal survives identically everywhere.

6 Usage

- `-p-twist R -twist-g57` — the full v2 path: all-g57 words, anchor-first seams, context consumption, slide, joint acceptance, unconditional birth-advance, ancestry union. Pure swap only (`-twist-neg-p` is ignored on this path).

- `TWIST_G57_NO_SLIDE=1` / `TWIST_G57_NO_RETRY=1` — env kill-switches for the two v2 placement features (A/B use).
- Omit `-twist-g57` for the legacy 3-CNOT swap-family packets (`-twist-neg-p` active there).
- `-db-advance` — separate flag; birth-advance for *DB splice* products. The g57 bracket path advances its insertions regardless.
- Report line: `twist-g57: consumed= emitted= net/seam[hist] solves= avg_us= slides= retries=`. Note: this path emits `ORIGIN_SYNT comp=1` material, so `comp=/shaped=` read population form under it (same caveat as `p_db>0`).
- Tuning constants (compile-time, `mix.rs`): `TG_SLIDE_CAP=512`, `TG_SLIDE_TRIES=3`, `TG_RETRIES=4`, `TG_ACCEPT_NET=+8`.

A The replacement table

The complete machine-generated table lives in `docs/G57_TWIST_REPLACEMENTS.txt`: **T1**, all 64 minimal (length-6) all-g57 words realizing $S = \text{swap}(a, b)$ (32 per helper wire; gate $[t, n, p]$ means $t \hat{=} n \vee \neg p$); **T2**, the 12 ordered pair-consumption identities on $\{a, b, c\}$ ($h_1 h_2 S \equiv R$ with $|R| = 4$, net +2); **T3**, the 8 single-gate attachment identities ($h S \equiv R$ with $|R| = 5$, net +4). The engine does not read this table — it re-derives everything from the BFS ball at startup — but the table is the exhaustive human-checkable inventory of what the seams can do at $k \leq 2$ on a 3-wire support. Representative entries:

kind	identity
bare word	$S \equiv [a, b, c] [a, c, b] [b, a, c] [b, c, a] [a, b, c] [a, c, b]$
pair, net +2	$[a, b, c] [a, c, b] S \equiv [b, a, c] [b, c, a] [a, b, c] [a, c, b]$
attach, net +4	$[a, b, c] S \equiv [a, c, b] [b, a, c] [b, c, a] [a, b, c] [a, c, b]$